E12He - RLC Resonant Circuits

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April 29, 2025

Abstract

This study examines the resonance curve and characteristics of two RLC series circuits with damping resistors R_d^1 =$(10 \pm 5\%)\Omega$ and R_d^2 = $(100 \pm 5\%)\Omega$ and a LC-Parallel R-Series circuit with damping resistor R_d^2 by fitting experimental data to theoretical impedance and voltage models. For the LC-Parallel R-Series circuit, measurements revealed a partial band-stop behavior centered at a fitted resonant frequency $f_{0,\text{fit}}$ =$(12.79 \pm 0.000)KHz$. Model fitting yielded quality factor Q =(0.0905 ± 0.0064), significantly lower than those of the RLC series configurations (Q =(0.1186 ± 0.0002) for R_d^1 and Q =(0.1128 ± 0.0002) for R_d^2), attributable to the high parallel impedance and additional damping from physical resistances and internal impedance of the measurement device. Fitted inductance and capacitance, values deviated from physical components by factors of $\approx 1/5$ and ≈ 4.5, respectively, reflecting the limitations of Thomson's undamped-circuit equation when applied to a damped, parallel circuit.

In the RLC Series circuit, both damping resistors (R_d^1 =$(10 \pm 5\%)\Omega$, R_d^2 =$(100 \pm 5\%)\Omega$) produced broad band-pass responses with tapering voltages. The fitted resonant frequencies ($f_{0,\text{fit}}^1$ =(10.652 ± 0.075)KHz, $f_{0,\text{fit}}^2$ =(11.128 ± 0.000085)KHz) were close yet distinct from the experimental values ($f_{0,\text{exp}}^1$ =(10.25 ± 0.125)KHz, $f_{0,\text{exp}}^2$ =$(11.75kHz \pm 0.125)$KHz), lying outside experimental uncertainties. Damping constants extracted from the fits (δ_{fit}^1 =$(273.652 \pm 0.015)KHz$, δ_{fit}^2 =$(336.645 \pm 0.022)KHz$) diverged from experimental values (δ_{exp}^1 =$(271.6 \pm 0.6)KHz$, δ_{exp}^2 =$(327.1 \pm 0.6)KHz$), though high coefficients of determination (r^2 =0.9961, r^2 =0.9984) confirm strong model correlation given correction for the inherent deviation caused by Thomson's equation. Experimental full-width at half-maximum (FWHM) data yielded $FWHM$ =(86.46 ± 0.18)KHz for R_d^1 and, by interpolation beyond the AWG's 100 kHz limit, $FWHM$ =$(104.13kHz \pm 0.18)$KHz for R_d^2. Calculated capacitances (C =$(13.53 \pm 0.07)\mu$ F, C =$(1.384 \pm 0.069)\mu$ F) reflecting the inverse proportionality to resistance inherent in the capacitance formula used.

Across both circuits, the implicit use of Thomson's undamped-oscillator equation introduced systematic discrepancies in extracted component values, in addition with unmodeled skin and cable effects, instrument impedance, and the Picoscope 2000A's internal resistance. For better accuracy, future work should employ damping-considered modeling, explicitly account for physical resistances in either the model used or within uncertainties, and expand data density around resonance peaks.

1 Introduction

This report examines the frequency-dependent voltage and current of an RLC Series resonant and LC Parallel - R Series resonant circuits, with the aim of exploring the resonance phenomenon in RLC circuits whilst gaining practical skills in circuit construction and analysis. Resonant circuits are electric circuits that produce or respond to resonance, *(depending on specific configuration and input voltage)*.

A key function of a resonant circuit is that it can act as a filter for selecting frequency ranges, and prevent particular frequencies from passing the circuit. Making the resonant circuit useful in electrical filters, *(band-pass/band-stop filters)*, producing electrical oscillations and in AC power systems to reduce reactive power and counter AC inductive effects, especially in variable AC power systems when impedance matching is not viable [1].

2 Theoretical Analysis

2.1 RLC Series Circuit

The key characteristics that determine the behavior of the output voltage/current and behavior of a resonant circuit are the circuits resonant frequency f_0 the impedance Z, the circuits quality factor Q, and the reactance of the Inductor and the capacitor X_L and X_C respectively. For this analysis, the output voltage is U_R which is measured across the resistor R_d, and the current through the circuit is [5]:

$$I = \frac{U_R}{R_d} \implies I \propto U_R \tag{1}$$

Where:

U_R : Voltage across the resistor R_d (V)
R_d : Resistance of R_d (Ω)

When analyzing the RLC Series Circuit, Fig. 1.

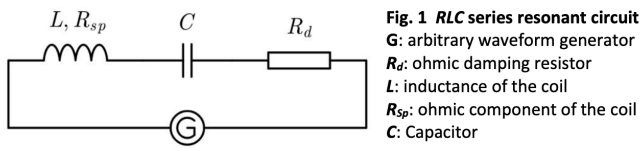

Figure 1: RLC Series Circuit Diagram

Source: [5]

At resonance, the imaginary part of the total impedance Z (see Eq.(4)) becomes zero, as the inductor reactance becomes equal to the capacitor reactance. Thus the resonant frequency of an RLC circuit can be found by equating $X_L = 2\pi fL$ and $X_C = \frac{1}{2\pi fC}$ to each other and solving for f, as f_0 is the point when these two reactances are equal

[4]:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{\omega_0}{2\pi} \tag{2}$$

Where:

L : Inductance of the Inductor (H)
C : Capacitance of the Capacitor (F)

The quality factor Q, is a measure of the energy efficiency of the circuit per oscillation cycle and is defined as [5]:

$$Q = \frac{f_0}{\Delta f} = \frac{\omega_0 L}{R_d + R_{sp}} \tag{3}$$

Where:

Δf : The Full-Width Half-Maximum, (FWHM) (Hz)
ω_0 : Angular frequency $\left(\frac{\text{rad}}{s}\right)$

A high Q value indicates little loss of energy, high Q resonant circuits are used in crystal filters and oscillators. A low Q value indicates significant loss of energy via dissipation, low Q resonant circuits have used as damping scenarios such as in constant voltage power supplies [2].

Examining the circuit in Fig.1 we can derive a formula for U_R by first considering the impedance Z, which is [5]:

$$Z = R_{sp} + R_d + i\,(X_L - X_C) = R_{\text{tot}} + i\big(\omega L - \frac{1}{\omega C}\big) \tag{4}$$

Then the transfer function $H(i\omega)$ and its amplitude $|H(i\omega)|$ can be evaluated as:

$$H(i\omega) = \frac{U_{\text{R}}}{U_0} = \frac{R_d}{Z} = \frac{R_d}{R_{\text{tot}} + i(\omega L - \frac{1}{\omega C})}$$

$$\implies |H(i\omega)| = \frac{R_d}{\sqrt{R_{\text{tot}}^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

And thus:

$$U_R = U_0 \frac{R_d}{\sqrt{R_{\text{tot}}^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \tag{5}$$

Another important characteristic to consider for a resonant circuit is the Full-Width at Half-Maximum, (FWHM) of the circuits resonance curve. Which is defined as, the width at half the height of the dissipated power:

$$P(f_1) = P(f_2) = \frac{P_{max}}{2} \implies \text{FWHM} = \Delta f = f_2 - f_1 \tag{6}$$

Where:

$P \quad = \frac{U_R^2}{R_d}$ is the power dissipated by R_d

This additional characteristic enables the introduction of a damping coefficient δ, with an experimental value of

$$\delta = \pi \Delta f \tag{7}$$

and theoretical value [6]:

$$\delta = \frac{R_{tot}}{2L} \tag{8}$$

As such with Eq.8 and U_R along with the simplification that $R_{tot} \approx R_d$:

$$U_R(\omega) = U_0 \frac{2\delta\omega}{\sqrt{(\omega^2 - \omega_0^2)^2 + (2\delta\omega)^2}} \quad : \quad \omega = 2\pi f \tag{9}$$

A model for the output voltage for a RLC series circuit is obtained.

2.2 LC-Parallel R-Series Circuit
(Derivation of Impedance Expression Task IV)

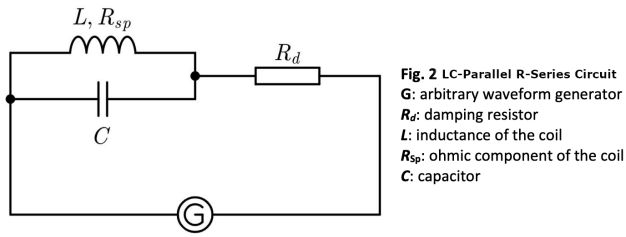

Figure 2: LC-Parallel R-Series Circuit Diagram
Source: [5]

To determine a model for the output voltage of a LC-Parallel R-Series Circuit, Fig.2, the transfer equation $H(i\omega)$:

$$H(i\omega) = \frac{U_{\text{R}}}{U_0} = \frac{R_d}{Z}$$

is again used however the total impedance Z of this circuit differs as the capacitor and the inductor are in parallel the total impedance for the combined components is:

$$Z_{LC} = \frac{1}{Z_L} + \frac{1}{Z_C} = \frac{Z_L Z_C}{Z_L + Z_C} \tag{10}$$

Drawing from the previous impedance derivation this results in the total impedance:

$$Z = R_d + Z_{\text{LC}}, \quad Z_{\text{LC}} = \frac{(R_{sp} + i\omega L) \cdot \frac{1}{i\omega C}}{(R_{sp} + i\omega L) + \frac{1}{i\omega C}}. \tag{11}$$

Finding the magnitude of the transfer equation $|H(i\omega)|$ and rearranging for U_R yields a model for the voltage across R_d for the circuit in Fig.2.

$$U_R = U_0 \frac{R_d}{\sqrt{(R_d + \text{Re}(Z))^2 + (\text{Im}(Z))^2}} \tag{12}$$

3 Experimental Analysis

3.1 RLC Series Circuit
(Task I & Task II)

3.1.1 Experimental Setup

The RLC Series Circuit was setup as seen in Fig.3 using a 0.22μF capacitor and 0.82mH inductor, for two resistors; $R_d^1 = (10 \pm 5\%)\Omega$ and $R_d^2 = (100 \pm 5\%)\Omega$. With the Picoscope 2000A acting as the AWG and the data collection device. The AWG voltage was set to 1V and the voltage across the resistor R_d was measured over a frequency range of 0.5 KHz to 100 KHz. With 0.5KHz intervals from (0.5 - 10)KHz then intervals of 0.25 KHz from (10.25 - 14)KHz followed by 0.5 KHz intervals from (14.5 - 61.5)KHz, and finally 1 KHz intervals from (62 -

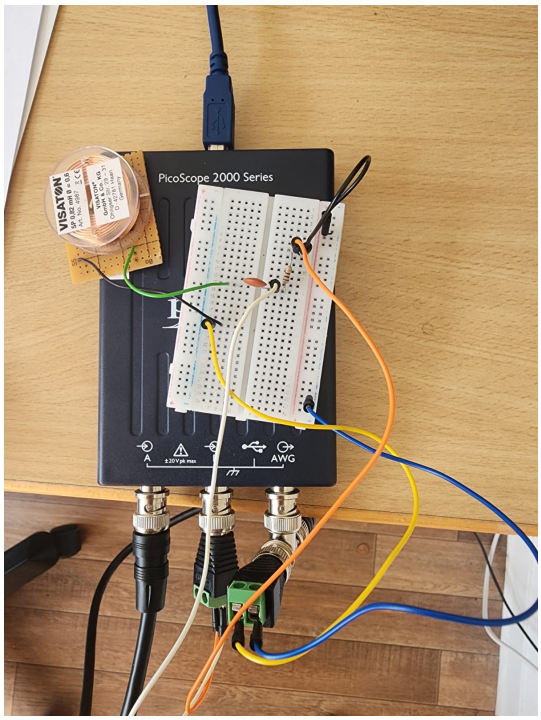

Figure 3: Experimental RLC Series Circuit

100)KHz. It was ensured that over this range the voltage U_R, remained within $0.2U_R(f_0) \leq U_R(f) \leq U_R(f_0)$ with consideration of uncertainty in experimental measurements.

Within the Picoscope 7 Software [3], the measurements were recorded using the in-built measurements feature using the 'Top' option *(Fig.4.4)* and recording the average value $\bar{x}$ *(Fig.4.1)* after $n \approx 1000$ samples *(Fig.4.2)*, the standard deviation σ *(Fig.4.5)* was also recorded for calculation of measurement uncertainty. The sampling rate of U_R was also adjusted to an '8 + 1' setup *(Fig.4.3)* to obtain a more stable σ reading.

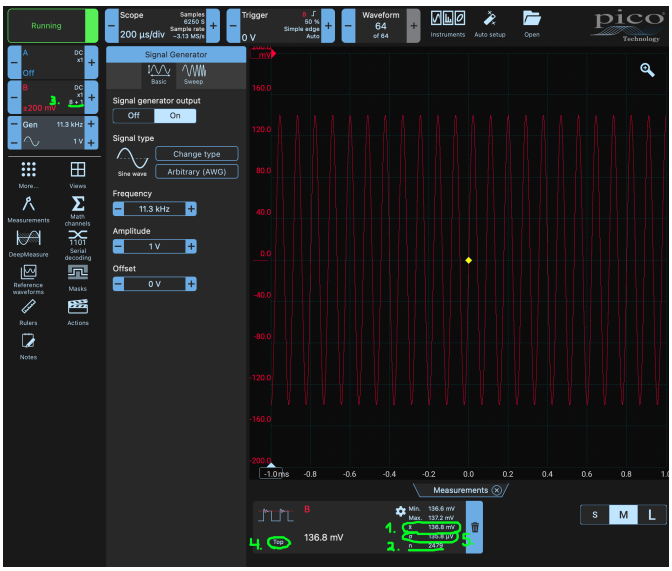

Figure 4: Picoscope 7 Main Panel

Both datasets for R_d^1 and R_d^2 were plotted on a resonance curve, and the FWHM, was determined and subsequently δ and Q were determined. Additionally the data was modeled to Eq.9 and theoretical values for f_0, δ and the Capacitance C were determined and compared with experimental findings. As for the uncertainty of each value collected, given each data point is a mean-value for ≈ 1000 samples at that particular f, the standard error of the mean *(SEM)* was used as the uncertainty for each value collected;

$$SEM = \frac{\sigma}{\sqrt{n}} \tag{13}$$

3.1.2 Data Analysis

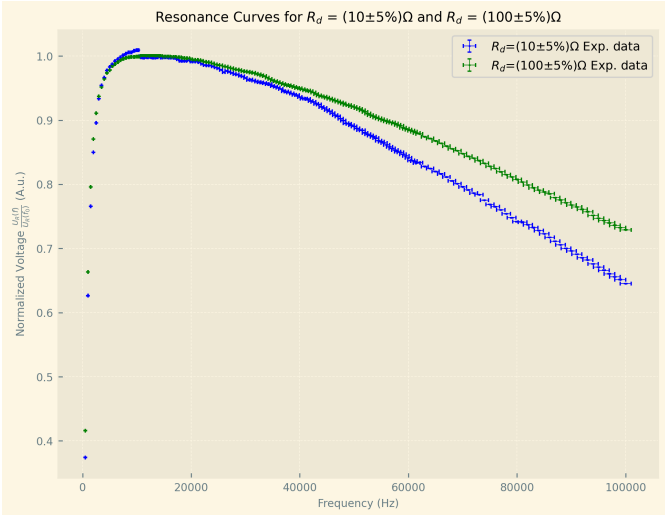

Figure 5: RLC Series Resonance Curve

Fig.5. directly compares the experiment data of R_d^1 and R_d^2, it is clear to see that the output voltage of the circuit U_R for $R_d^2 = (100 \pm 5\%)\Omega$ decays at a faster rate than that of R_d^1.

Fig.6. models the experimental data for both R_d^1 and R_d^2 through use of Eq.(9), along with a determination of the FWHM for each damping resistor. For the case of the 100Ω damping resistor the second intercept of the FWHM was outside of the operating range of the AWG, as such linear interpolation was used, based of a linear fit of the last 12 data points, to reach the second intercept. A linear fit was deemed an adequate solution as the tail-end of the experimental data is largely linear. It should be of note that additional uncertainty calculations between the experimental data and the linear fit were not made and the uncertainty of the second intercept for Eq.(6) was ultimately used from, the final data points of experimental data.

The fitted parameters $f_{0,\text{fit}}$ and δ_{fit} for each damping resistor was determined from the fitted RLC Curve with $f_{0,\text{fit}}$ for both damping resistors close to the experimental determination of $f_{0,\text{exp}}$ however outside the experimental uncertainty of $f_{0,\text{exp}}$, suggesting other factors may be the cause of the discrepancy.
The fitted damping coefficient δ_{fit} for R_d^1 and R_d^2 is $(273.652 \pm 0.015)KHz$ and and $(336.645 \pm 0.022)KHz$ respectively which, again is close to the experimental results δ_{exp} as outlined in Table.1, but outside of experimental

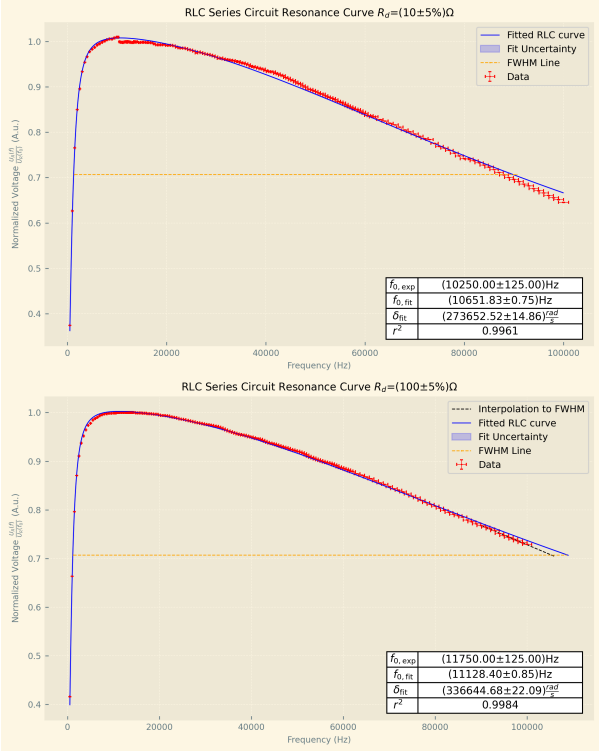

Figure 6: RLC Series Resonance Curve Analysis

Table 1: Metrics of RLC Circuits R_d^1 and R_d^2.

Parameter	**(10±5%)Ω (R_d^1)**	**(100±5%)Ω (R_d^2)**
FWHM (KHz)	86.46 ± 0.18	104.13 ± 0.18
Q	0.1186 ± 0.0002	0.1128 ± 0.0002
δ_{exp} (KHz)	271.6 ± 0.6	327.1 ± 0.6

and theoretical uncertainty further suggesting additional factors may be influencing the discrepancy. The fitted curve closely modeled the experimental data with an r^2 value for R_d^1 and R_d^2 of 0.9961 and 0.9984 respectively, showing a strong correlation between the model Eq.(9). and experimental findings.
From the determined values of FWHM, the quality factor Q of each circuit R_d^1,R_d^2 was also determined to be (0.1186 ± 0.0002) and (0.1128 ± 0.0002) respectively. Both are of similar but distinct values of ≈ 0.1150, which indicates significant energy loss across the circuit. Which is evident as the non-normalized resonant voltage $U_{R_d^1}(f_0) \approx 16.58$mV and $U_{R_d^2}(f_0) \approx 138$mV significantly less than the input voltage of 1V. The capacitance C was also computed across both circuits R_d^1 and R_d^2, given this was not changed between analysis the values should be within experimental uncertainty of each other, however for R_d^1, C_1 =(13.53±0.07)μ F and R_d^2, C_2 =(1.384±0.069)μ F differ by an order of magnitude, it is predicted that changing the resistor value R_d does not physically change the capacitance in the circuit. Given the capacitance was calculated using 2 and substituting L with Eq.8 rearranged, this discrepancy has been attributed to additional factors that were not considered.

3.1.3 Discussion

The general trend in data observed in both Fig.(5). and Fig.(6). was that of a band-pass circuit with a significantly wide band and a tapering/decaying voltage. The $R_d^2 = (100 \pm 5\%)\Omega$, resonance curve decayed at a faster rate than that of $R_d^1 = (10 \pm 5\%)\Omega$. Despite predictions using Thomson's Equation Eq.(2) the resonant frequency $f_{0,\text{fit}}$; $f_{0,\text{fit}}^1 = (10.652 \pm 0.075)$KHz,$f_{0,\text{fit}}^2 = (11.128 \pm 0.000085)$KHz, for both damping resistors R_d were close, but distinct and outside the uncertainty of their respective experimental values; $f_{0,\text{exp}}^1 = (10.25 \pm 0.125)$KHz,$f_{0,\text{exp}}^2 = (11.75kHz \pm 0.125)$KHz. This is most likely a result using Thomson's equation, which is for the particular case of an under-damped system, for the RLC series circuits with a damping resistor R_d.*(Reiterated in section 3.2.3)*.

It is believed that the equation:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \tag{14}$$

is more applicable to this circuit as it incorporates the effect of the damping resistor as well. However, Thomson's equation was implicitly used in the fitting model, Eq.(9). for U_R, it is suspected that this improper model for this scenario influences the discrepancy between the experimentally found damping constants for R_d^1, $\delta_{\text{exp}} = (271.6 \pm 0.6)KHz$ and R_d^2, $\delta_{\text{exp}} = (327.1 \pm 0.6)KHz$, from fitted values of R_d^1, $\delta_{\text{fit}} = (273.652 \pm 0.015)KHz$ and R_d^2, $\delta_{\text{fit}} = (336.645 \pm 0.022)KHz$, along with other factors such as internal impedance of the Picoscope 2000A Series used as the measuring device and AWG and other additional sources of resistance and impedance due to the real components used. Despite the oversight in utilizing Thomson's equation to determine the model equation, the correlation between the experimental data and the modeled parameters was strong for both resistors, with R_d^1 having $r^2 = 0.9961$ and R_d^2 having $r^2 = 0.9984$, indicating that the model used, if accounted for simplifications made, can accurately model experimental findings.

The $FWHM$ for the R_d^1 was determined to be $FWHM = (86.46 \pm 0.18)$KHz from experimental data. However, due to the 100KHz max operating range of the Picoscope 2000A Series AWG, the FWHM for the R_d^2 =$(100 \pm 5\%)\Omega$ could not be solely determined from experimental data. As a result, it was deemed sufficient to interpolate to the second intercept required for FWHM from the last 12 data points collected. This was deemed sufficient as the trend of the latter quarter of experimental data is largely linear and as such it is predicted that the $FWHM = (104.13kHz \pm 0.18)$KHz is deemed to be fairly accurate.

Using the determined values of δ_{fit} and $f_{0,\text{fit}}$ the capacitance was calculated using Eq.(2). and Eq.(8). for $R_d^1 = (10 \pm 5\%)\Omega$ this was determined as, $C = (13.53 \pm 0.07)\mu$ F and for $R_d^2 = (100 \pm 5\%)\Omega$, $C = (1.384 \pm 0.069)\mu$ F. It is initially curious that an increase in the damping resistor by a factor of 10, resulted in an inversely proportionally change in C, However considering the capacitance was determined from the Thomson Eq.(2). and Eq.(8), this dictates that $R \propto \frac{1}{C}$, thus inherently this relation will be present. When comparing these theoretical values to the physical component used; $C = 0.22\mu$F, it is evident that again the internal use of Thomson's equation has influenced the capacitance, however it is also suspected that the factors such as the internal resistance of cables, along with experimental

measurement through use of the Picoscope 2000A which itself has internal resistances, and impedance impact the results of C.

3.2 LC-Parallel R-Series Circuit
(Task III & Task IV)

3.2.1 Experimental Setup

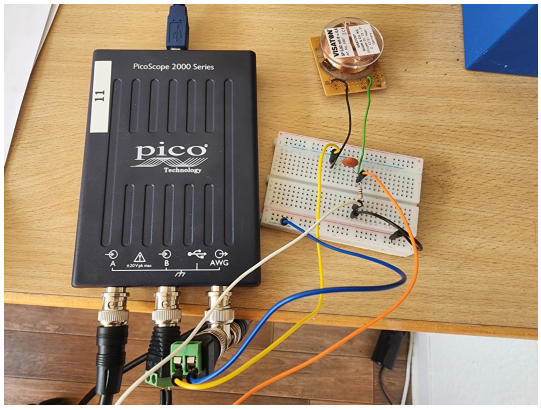

Figure 7: Experimental LC-Parallel R-Series Circuit

The LC-Parallel R-Series Circuit was setup as seen above in Fig.7 for a resistor $R = (100 \pm 5\%)\Omega$ and similar to the experimental setup and data collection of the RLC Series Circuit the Picoscope 2000A acted as both AWG and data collection device, the AWG voltage was set to 1V, and frequency varied from 0.5 KHz to 100 KHz with intervals of 0.5 KHz one the frequency range (0.5 - 10)KHz, then intervals of 0.25 KHz from (10.25 - 15.5)KHz followed by intervals of 0.5 KHz from (16 - 37)KHz.

3.2.2 Data Analysis

As outlined in Section:2.2 the derivation impedance expression (11) is used with the transfer function $|H(i\omega)|$ to arrive at a model for the voltage $U_R(f)$ across the damping resistor R_d, Eq.12.

Fig.8. is the results of fitting that model to experimental data and shows a list of fitted parameters, along with Q and the experimentally determined resonant frequency $f_{0,\text{exp}}$.
Comparing the fitted resonant frequency $f_{0,\text{fit}} = (12.79 \pm 0.000)KHz$ and experimental determination from data collected, $f_{0,\text{exp}} = (12.50 \pm 0.125)KHz$ the results are extremely close, however still fall outside

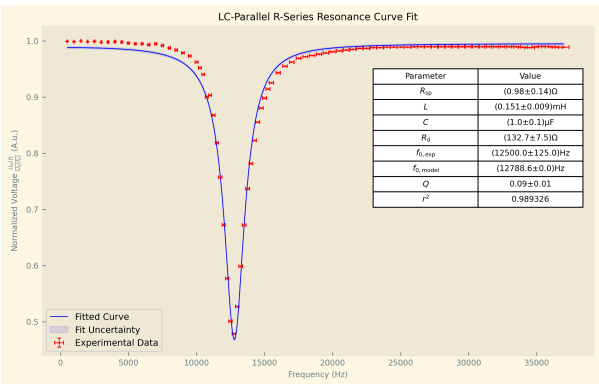

Figure 8: LC-Parallel R-Series Resonance Analysis

Table 2: Comparison of experimental measurements and fitted parameters

Parameter	Experimental Value	Fitted Value
f_0 (KHz)	12.50 ± 0.125	12.79 ± 0.000
R_d (Ω)	100 ± 5%	132.7 ± 7.5
R_{sp} (Ω)	NA	0.98 ± 0.14
L (mH)	0.82	0.151 ± 0.009
C (μF)	0.22	1.0 ± 0.01

of the uncertainty of $f_{0,\text{exp}}$. The internal resistance of the inductor R_{sp} was determined to be $(0.98 \pm 0.14)\Omega$, this resistance was not experimentally measured so it is uncertain as to the accuracy of this fitting parameter.

Comparing the remaining experimental values with fitted values, Table.2. It is clear that all fitted values are of the same order of magnitude, as their experimental counterpart but none fall within their respective experimental uncertainties. The fitted L and C differ significantly from their experimental counterparts with L being $\approx 0.2L_{\text{exp}}$ and $C \approx 4.5C_{\text{exp}}$. This discrepancy is predicted to be a result of unaccounted factors, but mainly with the complex modeling equation Eq.(12), which can result in difficulties in computing appropriate values for the parameters; L,C,R_d and R_{sp}, due to multiple possible solutions. For this fitting the parameters listed in Table.2. resulting in a strong correlation with experimental data, r^2 =0.9893 however it is uncertain whether this fitting and associated parameters are the most appropriate solution.

Additionally the quality factor Q was found to be (0.0905 ± 0.0064), which when comparing to the results of the RLC Series Circuit, $Q_1 = (0.1186 \pm 0.0002)$ and $Q_2 = (0.1128 \pm 0.0002)$ is significantly lower, suggesting that this LC-Parallel R-Series circuit is significantly less energy efficient than that of the RLC-Series circuits.

Comparing the fitted resonant frequency $f_0 = (12.79 \pm 0.000)KHz$ with the f_0 values determined from the RLC series circuit analysis of R_d^1, $f_0 = (10.652 \pm 0.075)$KHz and R_d^2, $f_0 = (11.128 \pm 0.000085)$KHz. It is expected that these values coincide within experimental uncertainty as dictated by Eq.(2). However it is evident that this is not the case. Each fitted value for f_0 are distinctly unique, further suggesting that there is an unaccounted aspect of the circuits not considered.

3.2.3 Discussion

The overall trend of Fig.8 conveys that this LC-Parallel R-Series circuit acts as a band-stop circuits with a relatively sharp band at its resonant frequency $f_{0,\text{fit}} = (12.79 \pm 0.000)KHz$ however the circuit does not act as a complete band-stop circuit as some voltage still passes the circuit. The discrepancy in fitted values is expected to be a result of simplification in the analysis performed on the circuit, aspects of the circuit such as skin-effect and internal resistance of cables are predicted to have impacted R_d and R_{sp}, along with experimental measurement through use of the Picoscope 2000A which itself has internal resistances, and impedance impact

the results of L. Additionally Thomson's Equation Eq.(2). was implicitly used in the derivation of the total impedance Z, (Eq.(11).), Thomson's formula is typically reserved for undamped circuits, given in the LC-Parallel R-Series circuit had a parallel component along with a damping resistor R_d =(100 ± 5%)Ω, this equation fails to model the relation between f_0, L and C and is predicted to contribute to the discrepancy in L and C. Additionally Thomson's equation involves a direct multiplication of L and C, given these are two undetermined parameters in the model equation Eq.(12), that are implicitly multiplied together resulting in difficulty distinguishing valid/appropriate values for the two, *(i.e. if LC = 5 then C=0.1, L=50 or C=1, L=5, etc.)*. This implicit multiplication is predicted to contribute to the $L \approx \frac{1}{5}L_{\text{exp}}$ and $C \approx 4.5C_{\text{exp}}$ discrepancies found in the fitted parameters.

Reviewing the comparison of the fitted resonant frequencies f_0 across the three different circuits. With the LC-Parallel R-Series circuit, $f_0 = (12.79 \pm 0.000)KHz$, and determined f_0 from the RLC series circuit analysis of R_d^1, $f_0 = (10.652 \pm 0.075)$KHz and R_d^2, $f_0 = (11.128 \pm 0.000085)$KHz. This simplification in Thomson's Equation also explains the differences from predictions to results from modeled data.

Assessing the quality factor $Q = (0.0905 \pm 0.0064)$ it is sensible that the value is low, as like the RLC series circuits, the circuit was constructed on a breadboard with many possible sources contributing to a loss of efficiency. Comparing Q from the LC-Parallel R-Series circuit with that of the RLC series Circuits $Q = (0.1186 \pm 0.0002)$ for R_d^1 and $Q = (0.1128 \pm 0.0002)$ for R_d^2, the significant reduction in Q for the LC-Parallel R-Series circuit can be explained by the LC-Parallel component and its Z_{LC} impedance, which for RLC-Parallel circuits is typically considerably larger, than in a series configuration, resulting in an increased loss of energy.

4 Conclusion

To conclude, the largest source of deviation for both the RLC-Series circuits and the LC-Parallel R-Series circuits was determined to be the use of Thomson's undamped-oscillator equation which incorrectly modeled the dynamics present in the circuits. This resulted in systematic deviations across all parameters determination via fitting. Additionally, for the LC-Parallel R-Series it became apparent that the use of complex models Eq.(9). and Eq.(12) proved difficult to find appropriate parameter values for fitting.

Analysis of fittings, showed that the RLC series circuits acted as band-pass filters with a tapering voltage as frequency increased beyond the resonance frequency,$f_{0,\text{fit}}^1$ =(10.652 ± 0.075)KHz for R_d^1 and $f_{0,\text{fit}}^2$ =(11.128 ± 0.000085)KHz for R_d^2. Whereas the LC-Parallel R-Series circuit acted as a band-stop filter around the fitted resonance frequency $f_{0,\text{fit}}$ =(12.79±0.000)KHz. Comparison of quality factor Q across the two types of circuits determined that while the RLC series circuits were largely energy inefficient with $Q_avg \approx 0.1150$ the LC-Parallel R-Series circuit was significantly more energy inefficient with $Q = (0.0905 \pm 0.0064)$. It is predicted that the quality factors were significantly low as a result of the circuits being constructed on a breadboard.

The determination of the capacitance across all circuits was unreliable due in part to unaccounted for physical effects such as the skin-effect and higher physical impedance's due to cable effects and internal impedance of the Picoscope 2000A used for measurements as well the use of Thomson's equation which directly multiplies L and C resulting in the determination of the capacitance for the LC-Parallel R-series circuit largely unreliable. It is recommended in future that models that consider damping, and are applicable for damping circuits be used, to explicitly account for physical resistances in either the model used or within uncertainties, and to record more data points around the resonance frequency as it largely dictates the values of all other fitted parameters.

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5 Appendix 1. Table of Components

(NOTE: Although the below tables in Appendix 1. and Appendix 2. are redundant they serve as reference material to values discussed and determined throughout the report.)

Table 3: Component List for circuits

Component	**(10±5%)Ω (R_d^1)**	**(100±5%)Ω (R_d^2)**	**LC-P R-S**
Capacitor,C (μF)	0.22	0.22	0.22
Inductor,L (mH)	0.82	0.82	0.82
Resistor,R_d (Ω)	$10 \pm 5\%$	$100 \pm 5\%$	$100 \pm 5\%$

6 Appendix 2. Table of Parameters/Metrics

Table 4: Parameters and Metrics for RLC circuit R_d^1 and R_d^2, and LC-P R-S circuits R_d^2.

Parameter/Metric	**(10±5%)Ω (R_d^1)**	**(100±5%)Ω (R_d^2)**	**LC-P R-S**
Capacitance,C (μF)	13.53 ± 0.07	1.384 ± 0.069	1.0 ± 0.01
Inductance,L (mH)	NA	NA	0.151 ± 0.009
Fitted Resistance,R_d (Ω)	NA	NA	132.7 ± 7.5
$f_{0,\text{exp}}$ (KHz)	10.25 ± 0.125	11.75 ± 0.125	12.50 ± 0.125
$f_{0,\text{theo}}$ (KHz)	10.652 ± 0.075	11.128 ± 0.000085	12.79 ± 0.000
Quality Factor Q	0.1186 ± 0.0002	0.1128 ± 0.0002	0.0905 ± 0.0064
Full Width at Half Maximum(FWHM) Δf (KHz)	86.46 ± 0.18	104.13 ± 0.18	NA
FWHM calculated Damping Coefficient δ_{exp} (KHz)	271.6 ± 0.6	327.1 ± 0.6	NA
Fitted Damping Coefficient δ_{fit}	273.652 ± 0.015	336.644 ± 0.022	NA
Inductor Resistance R_{sp}	NA	NA	0.98 ± 0.14